Positive shared-support mixed branch: independent UInt64 review

The composed theorem matches the retained positive, distinct-other branch at arithmetic-value scope. It starts from the actual outer word predicates: k=0, cancellation equality false, and scaled support coordinate distinct from p4.x. Canonical equality reflection transports those predicates into the field. Original normalized input validity supplies the positive ordinate equality through the reviewed field composition; no positive-equality or output-validity premise was added to the main word theorem.

The value flow is p4 from the mixed prelude, doubling of the original degree-one input, then generic addition(p4,doubled original). Operand order and four curve-coefficient positions match the Rust source. The intermediate point and doubled payload canonicality/decode are obtained before the generic word decoder is applied. Field output nonzero weights and validity are transported back through exact payload decoding. The original input word y is also proved nonzero. The literal-five boundary in doubling remains covered by its imported reviewed bridge for prime p=5.

positiveWordDispatch is intentionally restricted. Its None result outside positive/distinct support denotes exclusion from this theorem, not the behavior of the production complete dispatcher. In particular the reviewer checked a common case where this restricted dispatcher returns None but genericWordDispatch returns Some. Repeated-support tripling and cancellation outputs are excluded here. Under the theorem's literal guards, the restricted dispatcher returns Some with canonical output, nonzero weights and normalized Mumford validity.

Premises: prime 5<=p<2^31; ten canonical coordinate words; all four curve coefficients canonical; four nonzero input weights; both original normalized input-validity predicates on the common monic quintic with zero X4 coefficient; and the three actual outer word guards. Source fidelity also requires retained Field n=1, tables=None and ordinary execution with adequate counter headroom. There is no compiled Rust execution, operation/counter equivalence, divisor-class-sum or whole-walk claim. Pure intermediate value bundles may define values beyond early returns; no evaluation-trace equivalence is inferred.

Fresh private main module, producer nonunit F7 full-premise control, separate four-declaration axiom readback, and reviewer boundary controls all exited zero under the pinned toolchain. The reviewer controls cover p=5 cancellation precedence, the restricted-None/common-generic-Some distinction, repeated support exclusion and a coefficient equal to p failing canonicality. Readback lists only propext, Classical.choice and Quot.sound. The producer's two 120-second elaboration timeouts and other failed attempts remain preserved; they are tooling history, not mathematical counterexamples or performance measurements.

All 83 producer-manifest entries and 13 declared external source/dependency bindings matched before review; they were checked again after replay. Existing pinned imported dependency binaries were reused, not rebuilt transitively or validated by an independent binary checker. The reviewer authored prior WordGeneric and Doubling/WordDoubling components, each separately independently reviewed, but did not author this composition. Producer files were not modified.

Judgment: supported within the stated restricted word-value and conditional polynomial-validity scope, with no blocking discrepancy. Cost ratio remains null; no scientific trial or program-status decision was made.